

WEEK 12 — FORENSIC INVESTIGATION

PART C: Memory Analysis, Network & Investigation Conclusion

Cyberster Blue Team Internship — Week 12 — Days 88–91
Volatility 3 Memory Forensics | Process Tree | Network | Malfind | Conclusion

Case	M57.biz — Jo Insider Threat Investigation
RAM Image	jo-2009-11-16.mddramimage
System Time	2009-11-17 00:22:31 UTC (at acquisition)
OS (Memory)	Windows XP SP3 (32-bit)
Tool	Volatility 3
Acquisition	mdd_1.3.exe run from E:\RAM\ (removable media)
Focus	Running processes, command lines, malware scan, network, DLLs, TrueCrypt
Part C Verdict	No confirmed malware; suspicious soffice.bin → cmd.exe relationship identified

1. Memory Acquisition

Live memory acquisition was performed using mdd_1.3.exe (Memory DD v1.3) — a Windows memory dump utility. The prefetch artefact MDD_1.3.EXE-286446E8.pf confirms 4 executions with the last at 2009-11-25 00:26:05 UTC. Critically, mdd_1.3.exe was launched from **E:\RAM** — removable media — suggesting the acquisition tool was brought in on an external drive.

Command recovered from memory: mdd_1.3.exe -o joRam01.dmp

Field	Value
RAM Image	jo-2009-11-16.mddramimage
Tool	mdd_1.3.exe (Memory DD v1.3)
Command	mdd_1.3.exe -o joRam01.dmp
Launched From	E:\RAM\ (removable media)
Prefetch	MDD_1.3.EXE-286446E8.pf — run count: 4
Last Execute	2009-11-25 00:26:05 UTC
System Time	2009-11-17 00:22:31 UTC (at memory snapshot)
Analysis Tool	Volatility 3
OS (from memory)	Windows XP SP3 — 32-bit

Volatility commands executed:

windows.info
windows.pslist
windows.pstree
windows.cmdline
windows.malware.malfind
windows.dlllist
windows.truecrypt.Passphrase
windows.netscan (attempted)

```
Progress: 100.00      PDB scanning finished
Variable      Value
-----
Kernel Base   0x804d7000
DTB           0x39000
Symbols file:  C:\Program Files\Python311\Lib\site-packages\volatility3\symbols\windows\ntoskrnl.pdb\1B2D0DFE2FB94275
80615C9018E04692-2.json.xz
Is64Bit       False
IsPAE         False
Layer_name     0 WindowsIntel
memory_layer   1 FileLayer
KdDebuggerDataBlock 0x8054cde0
NTBuildLab    2600.xpsp_sp3_gdr.090804-1435
CSDVersion    3
KdVersionBlock 0x8054cdb8
Major/Minor   15.2600
MachineType   332
KeNumberProcessors 1
SystemTime    2009-11-17 00:22:31+00:00
NtSystemRoot  C:\WINDOWS
NtProductType NtProductWinNt
NtMajorVersion 5
NtMinorVersion 1
PE MajorOperatingSystemVersion 5
PE MinorOperatingSystemVersion 1
PE Machine    332
PE TimeDateStamp Tue Aug 4 15:14:34 2009
PE OS_C:\Windows\system32\
```

Figure 1 — Volatility 3 memory analysis output (windows.info / pslist)

2. Process Tree Analysis — windows.pstree

The process tree at the time of memory capture shows the following key processes. Firefox, Python, Java, and OpenOffice were all active — consistent with the disk-based findings of browser automation and steganography activity.

Process	Parent	Significance
explorer.exe	winlogon.exe	Windows shell — user session active
firefox.exe	explorer.exe	Browser active — targeted by patentauto.py via MozRepl port 4242
python.exe	explorer.exe	Python interpreter — patentauto.py possibly executing
javaw.exe	explorer.exe	Java runtime — diit-1.5.jar potentially active
soffice.bin	explorer.exe	OpenOffice — spawned multiple cmd.exe processes (suspicious)
cmd.exe	soffice.bin	Command prompt children of soffice.bin — unusual relationship
MSIMN.EXE	explorer.exe	Outlook Express — email client active at acquisition
mdd_1.3.exe	cmd.exe	Memory acquisition tool — launched from removable media
AVGCMGR.EXE	services.exe	AVG antivirus — run count 104 across prefetch

■ CAUTION

soffice.bin → cmd.exe (Critical): OpenOffice's binary spawned multiple cmd.exe processes. This is unusual behaviour and may indicate macro execution, scripted automation, or command execution through an OpenOffice macro. This finding is flagged for further analysis but is not confirmed as malicious.

```

Administrator: Windows PowerShell
\Device\HarddiskVolume1\Program Files\OpenOffice.org 3\program\soffice.bin "C:\Program Files\OpenOffice.org 3\program\soffice.exe" "-env:000_CMD=2C:\Program Files\OpenOffice.org 3\Basis\program" C:\Program Files\OpenOffice.org 3\program\soffice.bin
*** 3328 2392 cmd.exe 0x82cb1da0 0 - 0 False 2009-11-16 18:47:22.000000 UTC 2009-11-16 18:47:23.000000 UTC
*** 3968 2392 cmd.exe 0x826777a8 0 - 0 False 2009-11-16 18:47:21.000000 UTC 2009-11-16 18:47:22.000000 UTC
*** 1188 2392 cmd.exe 0x8295fda0 0 - 0 False 2009-11-16 18:47:22.000000 UTC 2009-11-16 18:47:22.000000 UTC
*** 2728 2392 cmd.exe 0x82f20da0 0 - 0 False 2009-11-16 18:47:21.000000 UTC 2009-11-16 18:47:21.000000 UTC
*** 3860 2392 cmd.exe 0x826d45c0 0 - 0 False 2009-11-16 18:47:16.000000 UTC 2009-11-16 18:47:20.000000 UTC
* 3996 472 firefox.exe 0x826dda0 0 - 0 False 2009-11-16 19:34:03.000000 UTC 2009-11-17 00:10:18.000000 UTC
* 3124 472 firefox.exe 0x826be020 0 - 0 False 2009-11-16 18:47:24.000000 UTC 2009-11-16 19:32:51.000000 UTC
* 856 472 jtsched.exe 0x82ccfb28 1 40 0 False 2009-11-13 04:49:20.000000 UTC N/A
\Device\HarddiskVolume1\Program Files\Java\jre6\bin\jtsched.exe "C:\Program Files\Java\jre6\bin\jtsched.exe" C:\Program Files\Java\jre6\bin\jtsched.exe
* 3096 472 cmd.exe 0x82735020 1 33 0 False 2009-11-17 00:21:37.000000 UTC N/A \Device\HarddiskVolume1\WINDOWS\system32\cmd.exe "C:\WINDOWS\system32\cmd.exe" C:\WINDOWS\system32\cmd.exe
* 3700 3096 mdd_1.3.exe 0x82f045d0 1 24 0 False 2009-11-17 00:22:29.000000 UTC N/A \Device\HarddiskVolume3\RAM\mdd_1.3.exe -o joRam01.dmp E:\RAM\mdd_1.3.exe
* 188 472 hkcmd.exe 0x82ef4910 2 100 0 False 2009-11-13 04:49:19.000000 UTC N/A \Device\HarddiskVolume1\WINDOWS\system32\hkcmd.exe "C:\WINDOWS\system32\hkcmd.exe" C:\WINDOWS\system32\hkcmd.exe
2148 3344 soffice.bin 0x82648da0 0 - 0 False 2009-11-17 00:08:10.000000 UTC 2009-11-17 00:08:16.000000 UTC \Device\HarddiskVolume1\Program Files\OpenOffice.org 3\program\soffice.bin
PS C:\Windows\system32>
  
```

Figure 2 — Process tree from Volatility 3 (pstree)

3. Command Line Analysis — windows.cmdline

CMD.EXE shows a prefetch run count of 11, confirming significant command-line activity. Volatility's cmdline plugin recovered the arguments passed to each process at the time of acquisition.

```

Administrator: Windows PowerShell
932 avgwdsvc.exe "C:\Program Files\AVG\AVG9\avgwdsvc.exe"
896 avgfws9.exe "C:\Program Files\AVG\AVG9\avgfws9.exe"
1384 jqs.exe "C:\Program Files\Java\jre6\bin\jqs.exe" -service -config "C:\Program Files\Java\jre6\lib\deploy\jqs\jqs
N c:\conf"
2376 avgemc.exe "C:\Program Files\AVG\AVG9\avgemc.exe"
2476 avgam.exe -
2540 avgnsx.exe -
2656 avgcsrnx.exe /pipeName=e942cc98-6619-4e62-8d57-0ce3e9eb97ab /coreSdkOptions=0 /binaryPath="C:\Program Files\
jec\AVG\AVG9\
3976 alg.exe C:\WINDOWS\System32\alg.exe
3120 avgcsrnx.exe /pipeName=6a228f79-575d-4a66-ad46-62cb86ddf05d /coreSdkOptions=18 /logConfFile="C:\Documents an
9 Settings\All Users\Application Data\avg9\temp\0a183012-1c5f-4320-baeb-c07e95394333-9ec-oopp.tmp" /loggerName=AVG.NS.Co
re /tempPath="C:\Documents and Settings\All Users\Application Data\avg9\temp\"
3180 avgcmgr.exe -
3656 soffice.exe "C:\Program Files\OpenOffice.org 3\program\soffice.exe"
2392 soffice.bin "C:\Program Files\OpenOffice.org 3\program\soffice.exe" "-env:OOO_CMD=C:\Program Files\OpenOf
fice.org 3\Basis\program"
3860 cmd.exe -
2728 cmd.exe -
3968 cmd.exe -
1188 cmd.exe -
3328 cmd.exe -
3124 firefox.exe -
3996 firefox.exe -
480 avgsrmax.exe -
3684 avgcmgr.exe -
2148 soffice.bin
3096 cmd.exe "C:\WINDOWS\system32\cmd.exe"
3780 mdd_1.3.exe mdd_1.3.exe -o joRam01.dmp
PS C:\WINDOWS\system32>

```

Figure 3 — windows.cmdline output: active process command lines

```

WARNING volatility3.plugins.windows.cmdscan: No conhost.exe processes found.
PS C:\Windows\system32> & "C:\Program Files\Python311\Scripts\vol.exe" -f "C:\Memory\jo-2009-11-16.mddramimage" windows.
consoles
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
WARNING volatility3.plugins.windows.consoles: No conhost.exe processes found.
PS C:\Windows\system32> & "C:\Program Files\Python311\Scripts\vol.exe" -f "C:\Memory\jo-2009-11-16.mddramimage" windows.
truecrypt.Passphrase
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
Offset Length Password
WARNING volatility3.plugins.windows.truecrypt: Truecrypt module not found in the modules list. Unable to proceed.
PS C:\Windows\system32> & "C:\Program Files\Python311\Scripts\vol.exe" -f "C:\Memory\jo-2009-11-16.mddramimage" windows.
malware.malfind
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
PID Process Start VPN End VPN Tag Protection CommitCharge PrivateMemory File output Notes
Hexdump Disasm
PID Process Start VPN End VPN Tag Protection CommitCharge PrivateMemory File output Notes
csrss.exe 0x7f6f0000 0x7f7effff Vad PAGE_EXECUTE_READWRITE 0 0 Disabled
N/A
08 00 00 00 cc 01 00 00 ff ee ff ee 08 70 00 00 .....p..
08 00 00 00 fe 00 00 00 10 00 00 20 00 00 .....
00 02 00 00 20 00 00 8d 01 00 00 ff ef fd 7f .....
03 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
03 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

```

Figure 4 — Additional command-line entries from memory

4. Malware Scan — windows.malware.malfind

The malfind plugin identifies memory regions with PAGE_EXECUTE_READWRITE permissions — a characteristic common in injected shellcode or dynamically generated executable code.

Finding	Associated Process	Assessment
Multiple PAGE_EXECUTE_READWRITE regions	soffice.bin	Suspicious but not confirmed malicious
Java-related memory strings	javaw.exe	Expected — Java JIT compilation creates executable memory
Dynamic executable memory	General	Requires deeper analysis; insufficient for malware confirmation
DLL injection	—	NOT confirmed
Reflective loading	—	NOT confirmed
Rootkit behaviour	—	NOT confirmed

■ CAUTION

Malfind flagged soffice.bin as having suspicious executable memory regions. However, this is insufficient to confirm malware. OpenOffice components legitimately use executable memory for rendering and macro engines. The soffice.bin → cmd.exe process relationship is more indicative of suspicious behaviour than the memory regions alone.

5. DLL Analysis — windows.dlllist

DLL Category	Processes	Assessment
Standard Windows XP DLLs	All processes	Expected — system libraries
AVG Antivirus modules	AVGCMGR.EXE and related	Expected — security software
Java libraries (jvm.dll etc.)	javaw.exe	Expected — needed for DIIT steganography tool
Networking DLLs (wsock32, ws2_32)	fox.exe, python.exe	Expected — network functionality
OpenOffice components	soffice.bin	Expected — office suite libraries
Hidden/injected DLLs	—	NOT identified

✓ ASSESSMENT

DLL analysis revealed no confirmed DLL injection, reflective loading, or hidden modules. All loaded libraries corresponded to known installed software. This does not exclude the possibility of in-memory execution without persistent DLLs.

6. TrueCrypt & Network Analysis

TrueCrypt Passphrase Analysis:

Plugin	Result	Interpretation
windows.truecrypt.Passphrase	No active passphrase recovered	TrueCrypt was not mounted at acquisition time
windows.netscan	Unsupported for WinXP SP3	No network reconstruction completed via Volatility

Network Evidence from Disk (corroborating):

- Firefox downloaded STU.doc from **noaa.gov**
- Firefox downloaded python-2.6.4.msi from **python.org**
- Firefox downloaded QuickTimeInstaller.exe from **apple.com CDN (edgesuite.net)**
- patentauto.py connected to **localhost:4242** (MozRepl) — local browser automation
- patentauto.py automated **USPTO patent website queries**

chatgpt.com/c/69cea371-2fc4-8320-8e98-64a0399088b8

Administrator: Windows PowerShell

```
Volatility 3 Framework 2.28.0
Progress: 100.00      PDB scanning finished
Offset Proto LocalAddr LocalPort ForeignAddr ForeignPort State PID Owner Created
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "C:\Program Files\Python311\Scripts\vol.exe\__main__.py", line 7, in <module>
  File "C:\Program Files\Python311\Lib\site-packages\volatility3\cli\__init__.py", line 934, in main
    CommandLine().run()
  File "C:\Program Files\Python311\Lib\site-packages\volatility3\cli\__init__.py", line 522, in run
    renderer.render(grid)
  File "C:\Program Files\Python311\Lib\site-packages\volatility3\cli\text_renderer.py", line 329, in render
    grid.populate(visitor, outfd)
  File "C:\Program Files\Python311\Lib\site-packages\volatility3\framework\renderers\__init__.py", line 318, in populate
    for level, item in self._generator:
  File "C:\Program Files\Python311\Lib\site-packages\volatility3\framework\plugins\windows\netscan.py", line 390, in _ge
generator
    netscan_symbol_table = self.create_netscan_symbol_table(
  File "C:\Program Files\Python311\Lib\site-packages\volatility3\framework\plugins\windows\netscan.py", line 347, in cre
ate_netscan_symbol_table
    symbol_filename, class_types = cls.determine_tcpip_version(
  File "C:\Program Files\Python311\Lib\site-packages\volatility3\framework\plugins\windows\netscan.py", line 318, in det
ermine_tcpip_version
    raise NotImplementedError(
NotImplementedError: This version of Windows is not supported: 5.1.15.2600!
```

Figure 5 — Network connection evidence from memory / disk analysis

Listing Keyword search 3 - zip x

Remote Monitoring Management Programs 2 Results

Source Name	S	C	O	Source Type	Score	Conclusion	Configuration	Justification	Category	File Path
system.LOG				File	Likely Notable		Remote Monitoring Management Programs		Kaseya (VSA)	/img_jo-2009
mstsc.exe				File	Likely Notable		Remote Monitoring Management Programs		Microsoft TSC	/img_jo-2009

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Figure 6 — Remote management programs identified on the system

7. Final Investigation Conclusion

Summary of Findings

Area	Key Finding	Risk
Patent Script	patentauto.py — automated Firefox-controlled patent intelligence collection	HIGH
Steganography	DIIT (diit-1.5.jar) + testHide.bmp — steganographic capability demonstrated	HIGH
Classified Doc	STU.doc from noaa.gov — US DoC SECRET-level briefing; no legitimate justification	HIGH
USB Activity	2 USB devices at suspicious moments; file duplication confirmed by MD5 hash	HIGH
File Concealment	Hidden/ folder for videos; files mirrored across USB images	MEDIUM
Memory — soffice	soffice.bin spawned cmd.exe processes — possible macro/automation execution	MEDIUM
Memory — Malfind	PAGE_EXECUTE_READWRITE in soffice.bin — suspicious, not confirmed malware	LOW
Command Line	11 CMD.EXE executions; mdd_1.3.exe run from removable media (RAM)	MEDIUM
DLL / Rootkit	No DLL injection, reflective loading, or rootkit behaviour identified	CLEAR
TrueCrypt	No active encrypted container at acquisition time	CLEAR
Network	netscan unsupported for WinXP; disk confirms external download of patent queries	MEDIUM

Analytical Conclusion

The forensic investigation of Jo's M57.biz workstation, USB images, and RAM capture presents a coherent and mutually corroborating pattern of suspicious insider activity. The evidence does not support a simple or innocent explanation.

On 2009-11-23, Jo installed Python 2.6.4 and immediately created patentauto.py — a purpose-built, technically sophisticated browser automation script that controls Firefox via MozRepl to perform structured, randomised, scheduled patent searches. An Alcor Micro USB flash drive was connected 51 seconds before the script was executed. On the same day, Jo accessed DIIT (a steganography tool) and testHide.bmp within 63 seconds of each other — consistent with a steganographic test operation. Binwalk found no payload, but DIIT-embedded data would require the original key to detect.

On 2009-11-24, a second unknown USB device was connected 4 seconds before STU.doc was accessed — a classified US Department of Commerce STU-III briefing. High-quality photographs were subsequently viewed. Memory analysis confirmed that Firefox, Python, Java, and OpenOffice were all active at the time of memory acquisition. OpenOffice's binary (soffice.bin) spawned multiple cmd.exe child processes — an unusual relationship suggesting macro or automation execution.

■ CRITICAL FINDING

Overall Verdict: The totality of evidence is consistent with a deliberate insider threat operation involving:

1. Systematic automated collection of patent intelligence using a custom Python/Firefox script
2. Demonstrated capability for steganographic data concealment (DIIT + testHide.bmp)
3. Unauthorised possession of a US government classified telecommunications document
4. Transfer of files to removable media — confirmed by MD5 hash duplication
5. Deliberate concealment of media files in a 'Hidden' subfolder

Memory analysis did not identify confirmed malware, rootkits, or DLL injection. The activity profile is consistent with a technically sophisticated user conducting intellectual property research and data exfiltration rather than a malware-compromised machine. The soffice.bin → cmd.exe relationship and memory malfind hits warrant further investigation before being discounted.

Recommended Next Steps

- Full forensic analysis of both USB images — deleted file recovery, full content inventory
- Re-examine testHide.bmp using DIIT itself to attempt payload extraction with likely keys
- Recover and analyse patentauto.py output data — any collected patent records on disk or USB
- Deep analysis of the soffice.bin → cmd.exe relationship — recover macro content from OpenOffice files
- Legal process for Jo's email accounts, external communication services, and internet history
- Full cmdline/cmdscan analysis to identify exact Python and DIIT commands executed
- Determine what patentauto.py was collecting — M57.biz patents or competitor intelligence?
- Clarify the business justification (if any) for STU.doc download from noaa.gov
- Investigate the contents of internalList.zip found in the AVG temp directory
- Serve legal process on M57.biz network logs for outbound connections on 2009-11-23 and 24